Zestaw 1

Zadanie 1. Działania oraz operacje na listach, podstawy

Definiowanie

Zdefiniować w notatniku dwie listy:

```
In[•]:= l1 = {a, b, c, d, e}
Out[•]= {a, b, c, d, e}

In[•]:= l2 = {1, 2, 3, 4, 5}
Out[•]= {1, 2, 3, 4, 5}
```

Wykonać następujące operacje na listach $l1$ oraz $l2$, a wyniki zapisać w notatniku:

Dodawanie, odejmowanie

```
In[•]:= l1 + 5
Out[•]= {5 + a, 5 + b, 5 + c, 5 + d, 5 + e}

In[•]:= l2 + 5
Out[•]= {6, 7, 8, 9, 10}

In[•]:= l1 + l2
Out[•]= {1 + a, 2 + b, 3 + c, 4 + d, 5 + e}

In[•]:= l1 - l2
Out[•]= {-1 + a, -2 + b, -3 + c, -4 + d, -5 + e}
```

Mnożenie, dzielenie

```
In[•]:= 5 * l1
Out[•]= {5 a, 5 b, 5 c, 5 d, 5 e}

In[•]:= 0 * l1
Out[•]= {0, 0, 0, 0, 0}

In[•]:= l1 * l2
Out[•]= {a, 2 b, 3 c, 4 d, 5 e}
```

In[•]:=
```
l2 / 0.5
```
Out[•]= {2., 4., 6., 8., 10.}

In[•]:=
```
l2 / (1 / 2)
```
Out[•]= {2, 4, 6, 8, 10}

In[•]:=
```
l1 / l2
```
Out[•]= $\left\{a, \frac{b}{2}, \frac{c}{3}, \frac{d}{4}, \frac{e}{5}\right\}$

Potęgowanie

In[•]:=
```
l1 ^ 5
```
Out[•]= $\{a^5, b^5, c^5, d^5, e^5\}$

In[•]:=
```
l2 ^ 5
```
Out[•]= {1, 32, 243, 1024, 3125}

In[•]:=
```
l1 ^ l2
```
Out[•]= $\{a, b^2, c^3, d^4, e^5\}$

In[•]:=
```
l1 ^ 0
```
Out[•]= {1, 1, 1, 1, 1}

Inne

In[•]:=

In[•]:=
```
l1.l2
```
Out[•]= a + 2 b + 3 c + 4 d + 5 e

In[•]:=
```
Sqrt[l1]
```
Out[•]= $\{\sqrt{a}, \sqrt{b}, \sqrt{c}, \sqrt{d}, \sqrt{e}\}$

In[•]:=
```
Sqrt[l2]
```
Out[•]= $\{1, \sqrt{2}, \sqrt{3}, 2, \sqrt{5}\}$

In[•]:=
```
Sqrt[N[l2]]
```
Out[•]= {1., 1.41421, 1.73205, 2., 2.23607}

In[•]:=
```
N[Sqrt[l2]]
```
Out[•]= {1., 1.41421, 1.73205, 2., 2.23607}

In[•]:=
```
Exp[l2]
```
Out[•]= $\{e, e^2, e^3, e^4, e^5\}$

In[•]:=
```
N[Exp[l2], 50]
```

Out[•]=
```
{2.71828182845904523536028747135266249775724709370000,
 7.38905609893065022723042746057500781318031557055180,
 20.08553692318766774092852965458171789698790783855400,
 54.59815003314423907811026120286087840279073703861400,
 148.41315910257660342111558004055227962348766759388}
```

Zadanie 2. Działania oraz operacje na listach, Total, Length

Dla list z *l1* oraz *l2* z poprzedniego zadania, obliczyć sumy: sum1 = Total[*l1*] oraz sum2 = Total[*l2*]

In[•]:=
```
sum1 = Total[l1]
```

Out[•]=
```
a + b + c + d + e
```

In[•]:=
```
sum2 = Total[l2]
```

Out[•]=
```
15
```

A następnie sprawdzić długość *l*1, *l*2, sum1 oraz sum2 przy użyciu funkcji Length[…].

In[•]:=
```
Map[Length, {l1, l2, sum1, sum2}]
```

Out[•]=
```
{5, 5, 5, 0}
```

Jaki był wynik? Zapisać w komentarzu wytłumaczenie, w tym celu zapoznać się z opisem funkcji Length[…].

In[•]:=
```
? Length
```

Out[•]=

Symbol

Length[*expr*] gives the number of elements in *expr*.

Find the number of terms in a polynomial:

In[•]:=
```
Length[1 + x + x^2 + x^9]
```

Out[•]=
```
4
```

Dla wielomianu, Length[…] zwraca liczę składników sumy

Zadanie 3. Działania oraz operacje na listach, Range

Wypisać listy zawierające następujące elementy korzystając z funkcji Range:

In[•]:= **? Range**

Out[•]=

Symbol

Range[i_{max}] generates the list {1, 2, …, i_{max}}.

Range[i_{min}, i_{max}] generates the list {i_{min}, …, i_{max}}.

Range[i_{min}, i_{max}, di] uses step di.

1) kolejne liczby naturalne od 1 do 25

In[•]:= **Range[1, 25]**

Out[•]= {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25}

2) kolejne liczby całkowite od –20 do 20

In[•]:= **Range[-20, 20]**

Out[•]= {-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}

3) liczby od 0 do 1, co 1/10

In[•]:= **Range[0, 1, 1/10]**

Out[•]= $\left\{0, \frac{1}{10}, \frac{1}{5}, \frac{3}{10}, \frac{2}{5}, \frac{1}{2}, \frac{3}{5}, \frac{7}{10}, \frac{4}{5}, \frac{9}{10}, 1\right\}$

4) liczby od 0 do 5, co 0.2

In[•]:=

Range[0, 5, 0.2]

Out[•]= {0., 0.2, 0.4, 0.6, 0.8, 1., 1.2, 1.4, 1.6, 1.8, 2., 2.2, 2.4, 2.6, 2.8, 3., 3.2, 3.4, 3.6, 3.8, 4., 4.2, 4.4, 4.6, 4.8, 5.}

In[•]:=

5) liczby od 10 do 0, co –2

In[•]:= **Range[10, 0, -2]**

Out[•]= {10, 8, 6, 4, 2, 0}

6) liczby od –2π do +2π, co π/4

```
In[•]:= Range[-2 Pi, 2 Pi, Pi / 4]
```

Out[•]= $\left\{-2\pi, -\frac{7\pi}{4}, -\frac{3\pi}{2}, -\frac{5\pi}{4}, -\pi, -\frac{3\pi}{4}, -\frac{\pi}{2}, -\frac{\pi}{4}, 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}, 2\pi\right\}$

7) liczby od a do 20a, co a

```
In[•]:= Range[a, 20 a, a]
```

```
Out[•]= {a, 2 a, 3 a, 4 a, 5 a, 6 a, 7 a, 8 a, 9 a, 10 a,
  11 a, 12 a, 13 a, 14 a, 15 a, 16 a, 17 a, 18 a, 19 a, 20 a}
```

Ciekawostka

Sprawdzić działanie funkcji Range[Range[10]], co uzyskujemy w wyniku?

```
In[•]:= Range[{1, 4, 2}]
```

```
Out[•]= {{1}, {1, 2, 3, 4}, {1, 2}}
```

```
In[•]:= Range[Range[10]]
```

```
Out[•]= {{1}, {1, 2}, {1, 2, 3}, {1, 2, 3, 4}, {1, 2, 3, 4, 5}, {1, 2, 3, 4, 5, 6}, {1, 2, 3, 4, 5, 6, 7},
  {1, 2, 3, 4, 5, 6, 7, 8}, {1, 2, 3, 4, 5, 6, 7, 8, 9}, {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}}
```

Zadanie 4. Działania oraz operacje na listach, Part, Take, Drop, Select, EvenQ, OddQ, PrimeQ oraz inne

Zdefiniować listę *l*3 będącą ciągiem kolejnych liczb naturalnych od 1 do 50 przy użyciu funkcji Range, a
następnie wybrać następujące elementy listy korzystając z zaproponowanych funkcji

```
In[•]:= l3 = Range[1, 50]
```

```
Out[•]= {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,
  17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
  34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50}
```

Funkcja Part, Take, Drop zapis

In[•]:= **? Part**

Out[•]=

Symbol

expr[[*i*]] or Part[*expr*, *i*] gives the ith part of *expr*.

expr[[−*i*]] counts from the end.

expr[[*i*, *j*, ...]] or Part[*expr*, *i*, *j*, ...] is equivalent to *expr*[[*i*]][[*j*]] ....

expr[[{i_1, i_2, ...}]] gives a list of the parts i_1, i_2, ... of *expr*.

expr[[*m* ;; *n*]] gives parts *m* through *n*.

expr[[*m* ;; *n* ;; *s*]] gives parts *m* through *n* in steps of *s*.

expr[["*key*"]] gives the value associated with the key "*key*" in an association *expr*.

expr[[Key[*k*]]] gives the value associated with an arbitrary key *k* in the association *expr*.

In[•]:= **? Take**

Out[•]=

Symbol

Take[*list*, *n*] gives the first *n* elements of *list*.

Take[*list*, −*n*] gives the last *n* elements of *list*.

Take[*list*, {*m*, *n*}] gives elements *m* through *n* of *list*.

Take[*list*, seq_1, seq_2, ...] gives a nested list in which elements specified by seq_i are taken at level *i* in *list*.

In[•]:= **? Drop**

Out[•]=

Symbol

Drop[*list*, *n*] gives *list* with its first *n* elements dropped.

Drop[*list*, −*n*] gives *list* with its last *n* elements dropped.

Drop[*list*, {*n*}] gives *list* with its nth element dropped.

Drop[*list*, {*m*, *n*}] gives *list* with elements *m* through *n* dropped.

Drop[*list*, {*m*, *n*, *s*}] gives *list* with elements *m* through *n* in steps of *s* dropped.

Drop[*list*, seq_1, seq_2, ...] gives a nested list in

which elements specified by seq_i have been dropped at level *i* in *list*.

1) pierwsze 10

```
In[•]:= Part[l3, ;; 10]
Out[•]= {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

In[•]:= l3[[ ;; 10]]
Out[•]= {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

In[•]:= Take[l3, 10]
Out[•]= {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

2) pierwszy, trzeci, piąty

```
In[•]:= l3[[{1, 3, 5}]]
Out[•]= {1, 3, 5}
```

3) ostatni

```
In[•]:= l3[[-1]]
Out[•]= 50
```

4) ostatnie 10

```
In[•]:= l3[[-10 ;;]]
Out[•]= {41, 42, 43, 44, 45, 46, 47, 48, 49, 50}

In[•]:= Take[l3, -10]
Out[•]= {41, 42, 43, 44, 45, 46, 47, 48, 49, 50}
```

5) bez pierwszych 20

```
In[•]:= l3[[20 + 1 ;;]]
Out[•]= {21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34,
          35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50}
```

In[•]:=
```
? Drop
```

Out[•]=

Symbol

Drop[*list*, *n*] gives *list* with its first *n* elements dropped.

Drop[*list*, –*n*] gives *list* with its last *n* elements dropped.

Drop[*list*, {*n*}] gives *list* with its nth element dropped.

Drop[*list*, {*m*, *n*}] gives *list* with elements *m* through *n* dropped.

Drop[*list*, {*m*, *n*, *s*}] gives *list* with elements *m* through *n* in steps of *s* dropped.

Drop[*list*, seq_1, seq_2, ...] gives a nested list in

which elements specified by seq_i have been dropped at level *i* in *list*.

In[•]:=
```
Drop[l3, 20]
```

Out[•]=
```
{21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34,
  35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50}
```

6) wszystkie nieparzyste

In[•]:=
```
l3[[ ;; ;; 2]]
```

Out[•]=
```
{1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49}
```

7) wszystkie parzyste

In[•]:=
```
l3[1 ;; ;; 2]
```

Out[•]=
```
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17,
  18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35,
  36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50}[1 ;; All ;; 2]
```

8) od 3-go elementu do 30-go elementu, co piąty

In[•]:=
```
l3[[3 ;; 30 ;; 5]]
```

Out[•]=
```
{3, 8, 13, 18, 23, 28}
```

Funkcja Select, wybierz wszystkie

In[•]:=
```
? Select
```

Out[•]=

Symbol

Select[*data*, *crit*] picks out all elements e_i of *data* for which *crit*[e_i] is True.

Select[*data*, *crit* → *prop*] returns the property *prop* of the selected elements.

Select[*data*, *crit*, *n*] picks out the first *n* elements for which *crit*[e_i] is True.

Select[*crit*] represents an operator form of Select that can be applied to an expression.

9) parzyste

In[•]:=
```
? Mod
```

Out[•]=

Symbol

Mod[*m*, *n*] gives the remainder on division of *m* by *n*.

Mod[*m*, *n*, *d*] uses an offset *d*.

In[•]:=
```
Select[l3, EvenQ]
```

Out[•]=
```
{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50}
```

10) nieparzyste

In[•]:=
```
Select[l3, OddQ]
```

Out[•]=
```
{1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49}
```

11) liczby pierwsze

In[•]:=
```
Select[l3, PrimeQ]
```

Out[•]=
```
{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47}
```

12) podzielne przez 7

In[•]:=
```
Select[l3, n ↦ Divisible[n, 7]]
```

Out[•]=
```
{7, 14, 21, 28, 35, 42, 49}
```

13) podzielne przez 3 oraz 7

```
In[•]:= Select[l3, n ↦ Divisible[n, 3] && Divisible[n, 7]]
Out[•]= {21, 42}
```

14) większe od 11.5

```
In[•]:= Select[l3, n ↦ n > 11.5]
Out[•]= {12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30,
  31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50}
```

15) wszystkie postaci $k^3 - 1$, gdzie $k \in \mathbb{N}$

```
In[•]:= Select[l3, n ↦ IntegerQ[CubeRoot[n + 1]]]
Out[•]= {7, 26}
```

16) takie, że silnia z elementu listy w zapisie dziesiętnym ma parzystą liczbę cyfr

```
In[•]:= ? IntegerLength
```

Out[•]=

Symbol

IntegerLength[n] gives the number of digits in the base 10 representation of the integer n.

IntegerLength[n, b] gives the number of digits in the base b representation of n.

```
In[•]:= Select[l3, n ↦ EvenQ[IntegerLength[n!, 10]]]
Out[•]= {4, 7, 9, 11, 13, 16, 18, 19, 21, 22, 24, 25, 28, 31, 32, 36, 37, 40, 41, 42, 46, 47, 48}
```

Zadanie 5. Rzut kostkami, Mean, StandardDeviation, Histogram, Count oraz inne

Zasymulować eksperyment z dwiema standardowym kostkami 6-ściennymi, A oraz B. Wygenerować używając funkcji `RandomInteger`, po n = 100 wyników rzutów dla kostki A oraz osobno dla kostki B.

In[•]:=
```
SeedRandom[1410]
```

Out[•]=
```
RandomGeneratorState[ Method: ExtendedCA
                      State hash: -4 098 803 455 797 552 762 ]
```

In[•]:=
```
? RandomInteger
```

Out[•]=

Symbol

RandomInteger[{i_{min}, i_{max}}] gives a pseudorandom integer in the range {i_{min}, i_{max}}.

RandomInteger[i_{max}] gives a pseudorandom integer in the range {0, …, i_{max}}.

RandomInteger[] pseudorandomly gives 0 or 1.

RandomInteger[*range*, n] gives a list of n pseudorandom integers.

RandomInteger[*range*, {n_1, n_2, …}] gives an $n_1 \times n_2 \times \ldots$ array of pseudorandom integers.

In[•]:=
```
RollTheDice = numberOfDices ↦ RandomInteger[{1, 6}, numberOfDices]
```

Out[•]=
```
Function[numberOfDices, RandomInteger[{1, 6}, numberOfDices]]
```

Wyniki rzutów zapisać odpowiednio jako lista lA oraz lista lB.

In[•]:=
```
lA = RollTheDice[100]
```

Out[•]=
```
{6, 6, 4, 6, 3, 1, 1, 6, 6, 5, 3, 6, 4, 4, 2, 5, 6, 3, 2, 4, 6, 3, 4,
 4, 1, 1, 5, 4, 1, 2, 1, 5, 4, 6, 5, 4, 2, 4, 3, 2, 4, 3, 6, 4, 4, 6, 4, 1,
 5, 6, 1, 4, 5, 3, 1, 6, 6, 3, 4, 1, 1, 1, 1, 6, 1, 5, 4, 5, 4, 4, 5, 1, 5, 4,
 6, 1, 5, 3, 4, 2, 1, 3, 4, 4, 1, 3, 2, 2, 6, 2, 4, 4, 1, 3, 1, 4, 1, 2, 6, 2}
```

In[•]:=
```
lB = RollTheDice[100]
```

Out[•]=
```
{3, 2, 3, 3, 2, 3, 5, 5, 6, 6, 5, 5, 1, 3, 3, 5, 3, 2, 5, 3, 4, 1, 6,
 3, 4, 5, 5, 3, 4, 4, 3, 5, 5, 1, 5, 3, 2, 1, 6, 3, 1, 1, 6, 5, 1, 1, 4, 1,
 4, 1, 1, 5, 2, 2, 5, 2, 6, 5, 5, 5, 5, 5, 6, 2, 4, 1, 1, 3, 6, 5, 3, 4, 6, 1,
 1, 2, 6, 4, 4, 6, 3, 1, 1, 3, 1, 1, 1, 5, 1, 3, 3, 2, 1, 6, 4, 5, 4, 5, 2, 5}
```

Zdefiniować listę zawierającą sumy wyników na obu kostkach w kolejnych rzutach, tj. $lAB = lB + lB$.

In[•]:=
```
lAB = lA + lB
```

Out[•]=
```
{9, 8, 7, 9, 5, 4, 6, 11, 12, 11, 8, 11, 5, 7, 5, 10, 9, 5, 7, 7, 10, 4, 10,
 7, 5, 6, 10, 7, 5, 6, 4, 10, 9, 7, 10, 7, 4, 5, 9, 5, 5, 4, 12, 9, 5, 7, 8, 2,
 9, 7, 2, 9, 7, 5, 6, 8, 12, 8, 9, 6, 6, 6, 7, 8, 5, 6, 5, 8, 10, 9, 8, 5, 11,
 5, 7, 3, 11, 7, 8, 8, 4, 4, 5, 7, 2, 4, 3, 7, 7, 5, 7, 6, 2, 9, 5, 9, 5, 7, 8, 7}
```

Następnie:

1) obliczyć wartość średnią z lAB

In[•]:=
```
Mean[N[lAB]]
```

Out[•]= 6.92

2) obliczyć odchylenie standardowe z lAB

$$\left(\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\hat{\mu})^2\right)^{1/2}$$

In[•]:=
```
StandardDeviation[N[lAB]]
```

Out[•]= 2.38971

3) sprawdzić jak wartość średnia z lAB zmienia się wraz ze wzrostem liczby rzutów n do $n = 10^4$ oraz $n = 10^6$.

Operować na wartościach numerycznych, jaka jest szacunkowa dokładność wyniku?

In[•]:=
```
multipleDiceRolls = Map[RollTheDice, {100, 10^4, 10^6}]
```

Out[•]=
```
{{3, 2, 1, 4, 6, 2, 2, 4, 6, 1, 4, 4, 1, 2, 6, 5, 4, 4, 2, 2, 2, 2, 1, 4, 3, 5, 4, 6, 1, 5, 1, 4, 1, 6, 3,
   5, 5, 5, 1, 6, 5, 1, 2, 4, 3, 1, 5, 6, 1, 6, 3, 3, 1, 2, 2, 1, 4, 6, 2, 5, 5, 2, 5, 4, 5, 2, 5, 6, 1,
   3, 5, 3, 2, 2, 1, 5, 5, 2, 3, 5, 3, 6, 1, 4, 3, 1, 2, 6, 3, 5, 2, 2, 5, 5, 3, 2, 6, 5, 6, 6}, ··· 1 ··· ,
 {3, 3, 5, 1, 3, 5, 6, 6, 1, 6, 2, 5, 3, 4, 6, 6, 6, 4, 3, 2, 3, 1, 2, 4, 4, 1, 5, 1, 4, 6, 4, 6, 1, 4, 6, 6,
   3, 2, 1, 6, 2, 4, 5, 1, 4, 4, 4, 2, 4, 3, 2, 1, 6, 6, 4, 1, 4, 4, 2, 3, 4, 5, 2, 4, 5, 2, 3, 6, 4, 6, 5, 1, 4,
   2, 1, 3, 3, 1, 5, 1, 6, 1, 5, 3, 3, 6, 5, 3, 1, 4, 6, 3, 1, 6, 6, 3, 2, 1, 1, 4, 4, 6, 2, 5, 1, 5, 2, 5, 5, 3,
   5, 1, 1, 1, 2, 4, 5, 1, 5, 4, 2, 3, 4, 4, 5, 1, 4, 3, 2, 3, 2, 2, 1, 1, 4, 6, 4, 6, 3, 3, 3, 1, 1, 5, 4, 2,
   1, 1, 6, 5, 6, 1, 4, 2, 6, 1, 4, 2, 4, 5, 6, 2, 3, 2, 6, 2, 1, 3, 3, 2, 4, 2, 3, 4, 4, 3, 1, 3, 3, 5, 2, 1,
   4, 5, 5, 4, 3, 2, 1, 5, 4, 2, 1, 6, 3, 4, 3, ··· 999 607 ··· , 1, 3, 1, 2, 3, 6, 5, 2, 3, 5, 5, 5, 2, 2, 5, 1,
   4, 3, 3, 6, 5, 4, 4, 2, 5, 1, 3, 3, 6, 1, 1, 4, 5, 2, 4, 1, 6, 2, 4, 3, 5, 5, 3, 4, 5, 3, 2, 1, 4, 4, 5, 4,
   6, 4, 6, 4, 3, 1, 2, 3, 6, 3, 3, 3, 2, 5, 1, 4, 5, 2, 2, 4, 6, 2, 3, 3, 2, 2, 1, 2, 5, 6, 3, 5, 5, 2, 4, 1,
   2, 6, 3, 4, 3, 6, 6, 2, 6, 6, 1, 2, 6, 6, 1, 2, 2, 2, 1, 5, 2, 4, 6, 6, 1, 3, 6, 3, 6, 4, 2, 1, 4, 3, 3, 4,
   2, 6, 6, 1, 1, 3, 6, 5, 2, 1, 6, 3, 4, 6, 5, 4, 2, 4, 1, 2, 5, 2, 6, 1, 6, 6, 6, 3, 5, 3, 3, 2, 4, 3, 4, 4,
   5, 1, 6, 2, 6, 6, 3, 3, 5, 2, 2, 4, 6, 4, 2, 1, 6, 2, 6, 4, 2, 6, 6, 3, 6, 1, 4, 2, 2, 1, 2, 5, 3, 6, 3, 5}}
```

Full expression not available (original memory size: 8.1 MB)

In[•]:=
```
Map[rolls ↦ Mean[N[rolls]], multipleDiceRolls]
```

Out[•]= {3.45, 3.4883, 3.49934}

Porównać wynik z wartością odchylenia standardowego z lAB podzieloną przez $\sqrt{n}$.

In[•]:=
```
Map[rolls ↦ StandardDeviation[N[rolls]] / Sqrt[Length[rolls]], multipleDiceRolls]
```

Out[•]= {0.174295, 0.0171481, 0.00170694}

4) dla $n = 10^4$ przedstawić wartości lAB na wykresie tak by można było odczytać prawdopodobieństwo uzyskania poszczególnych wartości, 2, 3, ..., 12.

In[]:=
```
n = 10^4
```

Out[]=
```
10 000
```

In[]:=
```
lAB = RollTheDice[n] + RollTheDice[n]
```

Out[]=
```
{4, 8, 6, 12, 2, 5, 2, 9, 4, 8, 10, 7, 7, 10, 7, 7, 3, 10, 10, 7, 6, 6, 4, 8, 8, 7, 6, 12, 8, 6, 12, 6, 4, 6,
 8, 7, 9, 9, 5, 3, 5, 6, 7, 7, 9, 7, 12, 3, 8, 6, 7, 8, 3, 5, 5, 10, 5, 8, 9, 6, 8, 6, 9, 3, 7, 7, 9, 6, 3,
 9, 6, 9, 3, 8, 7, 6, 2, 6, 3, 8, 5, 2, 7, 8, 7, 8, 4, 8, 6, 7, 10, 7, 6, 5, 8, 4, 7, 9, 12, 8, 7, 9, 10, 5,
 6, 7, 5, 6, 7, 11, 7, 9, 5, 6, 10, 4, 2, 9, 5, 7, 6, 11, 5, 4, 8, 8, 5, 6, 5, 7, 10, 5, 5, 12, 10, 9, 10,
 9, 7, 6, 4, 7, 8, 4, 3, 10, 7, 4, 9, 6, 9, 6, 8, 12, 9, 9, 5, 7, 11, 5, 4, 8, 7, 8, 9, 2, 6, 4, 9, 12, 6,
 9, 7, 9, 9, 7, 7, 9, 12, 8, 7, 7, 7, 4, 3, 8, 4, 8, 3, 7, 10, 5, 9, 7, 9, 6, 7, 8, 7, 9, 7, 3, 5, 5, 9, 4,
 7, 10, 10, 4, 8, 8, 6, 9, 8, 10, 6, 5, 7, 9, 5, 7, 6, 10, 7, 9, 8, 7, 7, 3, 10, 8, 3, 2, 5, ··· 9531 ···,
 6, 9, 5, 11, 6, 7, 10, 7, 8, 8, 8, 7, 5, 4, 9, 8, 12, 8, 10, 5, 6, 8, 8, 7, 4, 10, 5, 7, 8, 3, 6, 7, 8, 5,
 12, 9, 9, 4, 9, 7, 7, 9, 5, 10, 5, 5, 7, 7, 4, 4, 4, 3, 11, 3, 12, 4, 9, 3, 12, 8, 10, 5, 4, 6, 11, 8, 7,
 7, 2, 8, 9, 8, 6, 5, 8, 5, 7, 9, 6, 5, 12, 8, 4, 5, 9, 10, 9, 6, 11, 7, 6, 11, 4, 9, 6, 6, 7, 10, 6, 6,
 6, 8, 8, 2, 8, 9, 9, 9, 4, 10, 8, 5, 9, 8, 9, 9, 10, 4, 10, 7, 4, 9, 7, 7, 7, 7, 4, 11, 8, 7, 3, 9, 10,
 11, 7, 8, 6, 4, 9, 9, 8, 9, 7, 4, 9, 7, 6, 8, 11, 4, 4, 2, 6, 7, 8, 8, 6, 9, 6, 8, 8, 10, 11, 7, 2, 2, 8,
 8, 6, 6, 5, 8, 11, 8, 2, 6, 5, 12, 7, 8, 5, 10, 7, 9, 6, 6, 5, 3, 9, 10, 5, 7, 8, 6, 3, 8, 6, 6, 6, 11,
 7, 4, 4, 7, 5, 5, 8, 7, 7, 4, 11, 9, 5, 8, 4, 12, 5, 7, 6, 8, 7, 9, 7, 6, 8, 10, 2, 8, 9, 8, 8, 7, 7, 7}
```

Full expression not available (original memory size: 80.2 kB)

In[]:=
```
counts = Counts[lAB]
```

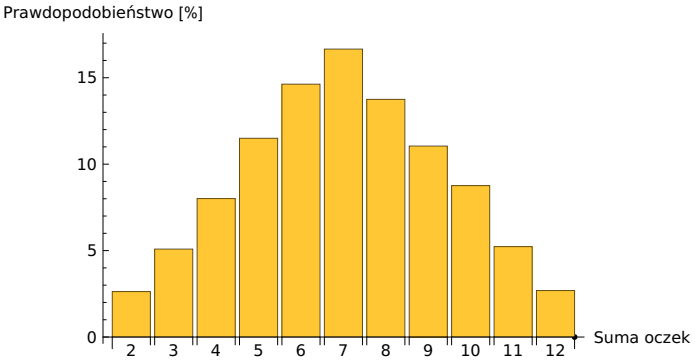
Out[]=
```
⟨|4 → 801, 8 → 1375, 6 → 1463, 12 → 269, 2 → 263,
 5 → 1150, 9 → 1105, 10 → 876, 7 → 1666, 3 → 509, 11 → 523|⟩
```

In[]:=
```
keys = Sort[Keys[counts]]
```

Out[]=
```
{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}
```

In[]:=
```
BarChart[
 Map[key ↦ Labeled[100 * N[counts[key]] / n, key], keys],
 AxesLabel → {"Suma oczek", "Prawdopodobieństwo [%]"}
]
```

Out[]=



5) jaka jest szansa (prawdopodobieństwo) na wyrzucenie:

Wskazówka: zapoznaj się z funkcją `Count` albo `BinCounts`

```
In[•]:= DiceCountToProbability = count ↦ PercentForm[N[count] / n]
```

```
Out[•]= Function[count, N[count]/n]
```

a) 2

```
In[•]:= DiceCountToProbability[Count[lAB, 2]]
```

```
Out[•]//PercentForm=
2.63%
```

b) 7

```
In[•]:= DiceCountToProbability[Count[lAB, 7]]
```

```
Out[•]//PercentForm=
16.66%
```

c) 6 albo 8

```
In[•]:= counts = Counts[lAB]
```

```
Out[•]= ⟨|4 → 801, 8 → 1375, 6 → 1463, 12 → 269, 2 → 263,
  5 → 1150, 9 → 1105, 10 → 876, 7 → 1666, 3 → 509, 11 → 523|⟩
```

```
In[•]:= DiceCountToProbability[counts[6] + counts[8]]
```

```
Out[•]//PercentForm=
28.38%
```

d) liczby nieparzystej

```
In[•]:= DiceCountToProbability[Total[
   Map[key ↦ counts[key], Select[keys, OddQ]]
  ]]
```

```
Out[•]//PercentForm=
49.53%
```

e) liczby parzystej

```
In[•]:= DiceCountToProbability[Total[
          Map[key ↦ counts[key], Select[keys, EvenQ]]
        ]]
Out[•]//PercentForm=
        50.47%
```

6) ile wynoszą powyższe prawdopodobieństwa obliczone teoretycznie?

Można rozwiązać zadanie „na kartce” a wyniki wpisać do notatnika albo wspomóc się programem Mathematica®.

```
In[•]:= diceResults = Range[6]
Out[•]= {1, 2, 3, 4, 5, 6}

In[•]:= dicePairs = Tuples[{diceResults, diceResults}]
Out[•]= {{1, 1}, {1, 2}, {1, 3}, {1, 4}, {1, 5}, {1, 6}, {2, 1}, {2, 2}, {2, 3}, {2, 4}, {2, 5}, {2, 6},
         {3, 1}, {3, 2}, {3, 3}, {3, 4}, {3, 5}, {3, 6}, {4, 1}, {4, 2}, {4, 3}, {4, 4}, {4, 5}, {4, 6},
         {5, 1}, {5, 2}, {5, 3}, {5, 4}, {5, 5}, {5, 6}, {6, 1}, {6, 2}, {6, 3}, {6, 4}, {6, 5}, {6, 6}}

In[•]:= diceSums = Map[pair ↦ Total[pair], dicePairs]
Out[•]= {2, 3, 4, 5, 6, 7, 3, 4, 5, 6, 7, 8, 4, 5, 6, 7, 8,
         9, 5, 6, 7, 8, 9, 10, 6, 7, 8, 9, 10, 11, 7, 8, 9, 10, 11, 12}

In[•]:= sumCounts = Counts[diceSums]
Out[•]= ⟨|2 → 1, 3 → 2, 4 → 3, 5 → 4, 6 → 5, 7 → 6, 8 → 5, 9 → 4, 10 → 3, 11 → 2, 12 → 1|⟩

In[•]:= probabilities = Map[i ↦ i / Length[dicePairs], sumCounts]
Out[•]= ⟨|2 → 1/36, 3 → 1/18, 4 → 1/12, 5 → 1/9, 6 → 5/36, 7 → 1/6, 8 → 5/36, 9 → 1/9, 10 → 1/12, 11 → 1/18, 12 → 1/36|⟩

In[•]:= N[probabilities]
Out[•]= ⟨|2 → 0.0277778, 3 → 0.0555556, 4 → 0.0833333, 5 → 0.111111, 6 → 0.138889, 7 → 0.166667,
          8 → 0.138889, 9 → 0.111111, 10 → 0.0833333, 11 → 0.0555556, 12 → 0.0277778|⟩
```

Jaka jest szansa (prawdopodobieństwo) na wyrzucenie:

a) 2

```
In[•]:= result = probabilities[2]
Out[•]= 1/36
```

```
In[•]:= PercentForm[N[result]]
```

Out[•]//PercentForm=

2.778%

b) 7

```
In[•]:= result = probabilities[7]
```

$$\text{Out}[\bullet]= \frac{1}{6}$$

```
In[•]:= PercentForm[N[result]]
```

Out[•]//PercentForm=

16.67%

c) 6 albo 8

```
In[•]:= result = probabilities[6] + probabilities[8]
```

$$\text{Out}[\bullet]= \frac{5}{18}$$

```
In[•]:= PercentForm[N[result]]
```

Out[•]//PercentForm=

27.78%

d) liczby nieparzystej

```
In[•]:= result = Total[
          Map[key ↦ probabilities[key], Select[keys, OddQ]]
        ]
```

$$\text{Out}[\bullet]= \frac{1}{2}$$

e) liczby parzystej

```
In[•]:= result = Total[
          Map[key ↦ probabilities[key], Select[keys, EvenQ]]
        ]
```

$$\text{Out}[\bullet]= \frac{1}{2}$$